PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

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Program / Batch	Cybersecurity Internship – B084 (NetworkWalks)
Date	12 November 2026
Modules Completed	W2-PM1 – Multiple Kali Tools; W2-PM5 – Zenmap Scanning
Client / Target	networkwalks.com (written permission stated as secured); own local LAN
Permission Secured?	Yes, for the stated authorised scope
Phases Covered	Phase 1: Reconnaissance & Footprinting; Phase 2: Scanning & Network Discovery; Phases 3–5: In progress

1. LIABILITY DISCLAIMER

I performed the activities described in this report only against systems for which permission was stated to have been secured, or devices and networks that I own. This report is prepared for cybersecurity education and research. It must not be used to facilitate unauthorised access, disruption, data collection or other unlawful activity.

The author, instructors and NetworkWalks are not responsible for misuse of the information presented. The person conducting any security activity is responsible for ensuring that the activity is lawful, properly authorised and within the agreed scope. Unauthorised access may result in criminal penalties, financial liability, employment consequences and other sanctions.

This report records reconnaissance and network-discovery exercises. It does not claim that exploitation, privilege escalation, persistence or destructive testing was performed.

2. INTRODUCTION

This report documents Week 2 practical activities completed as part of the NetworkWalks Cybersecurity Internship. It covers footprinting and reconnaissance of the networkwalks.com domain using Kali Linux tools under the stated written authorisation, together with network discovery of the author's own local network using Zenmap.

The two modules demonstrate complementary stages of an assessment. Reconnaissance collects publicly available information about a domain and its web-facing services. Network discovery identifies responding hosts within an authorised local network and helps build an inventory of visible devices.

The footprinting exercises used WHOIS, WhatWeb, Nslookup, cURL, WAFW00F and DNSRecon. Zenmap (the graphical interface for Nmap) was used for the local network discovery exercise. Windows command-line tools were used to identify the local network configuration before scanning.

The findings below are limited to the observations supplied for this report. Where an exact scan output, screenshot, timestamp or local network address was not supplied, the report identifies that limitation rather than inventing evidence.

3. TOOLS USED

Tool	Purpose
Kali Linux	Operating system used for reconnaissance exercises.
Windows	Host operating system used for local network information and Zenmap activities.
WHOIS	Retrieve publicly available domain registration and name-server information.
WhatWeb	Identify technologies and web components observable from the website.
Nslookup	Query DNS to resolve domain names and inspect DNS

	responses.
cURL (-I)	Request and inspect HTTP response headers.
WAFW00F	Identify indicators of a Web Application Firewall.
DNSRecon	Collect and review DNS-related records.
Zenmap (Nmap GUI)	Perform authorised host discovery and review scan results and topology.
Windows CMD (ipconfig)	Identify the local host's IP configuration and network details.

4. ACTIVITIES PERFORMED

4.1 Footprinting and Reconnaissance

Reconnaissance was performed against networkwalks.com using six Kali Linux tools. Each tool provided a different view of the publicly observable domain or web service.

WHOIS was used to review publicly available domain registration information, including registration-related details and name-server information. This information can help an assessor understand the domain's administrative and DNS context.

WhatWeb was used to fingerprint web technologies. The supplied observation identified WordPress 7.0.4 and WP Download Manager 3.3.58. These are technology-identification observations only; they do not establish that either component is vulnerable.

Nslookup was used to resolve the domain. The supplied result identified the IPv4 address 192.232.216.135. DNS answers can vary by time, resolver, geographic location and hosting configuration, so this value should be treated as the result observed during the exercise rather than a permanent address.

cURL was used with the -I option to inspect HTTP response headers. The supplied notes state that the response exposed the /wp-json/ WordPress REST API endpoint. The presence of a standard API endpoint is not, by itself, evidence of a security weakness.

WAFW00F was used to check for indications of a Web Application Firewall. The supplied result identified ModSecurity (SpiderLabs). This indicates a WAF-related technology signature was detected; it does not prove that all relevant traffic is inspected or that the WAF blocks every attack.

DNSRecon was used to review DNS information, including name-server, mail-server, SPF/TXT and service-related records. Specific record values were not included in the supplied notes and are therefore not reproduced here.

4.2 Network Scanning and Discovery with Zenmap

The second activity used Zenmap to discover responding hosts on the author's own local network. Windows ipconfig was used first to identify the local IP configuration and subnet. The relevant authorised subnet was then entered into Zenmap and the Ping Scan profile was selected for host discovery.

The supplied practical notes list four example responding addresses: 10.0.0.1, 10.0.0.4, 10.0.0.19 and 10.0.0.5. However, the notes explicitly describe these as example results and do not establish that they were the final results from the author's own LAN. They are therefore illustrative training values, not verified assets.

The practical also involved reviewing host IP and MAC address information, interpreting host-discovery results, opening Zenmap's Topology view, enabling the legend and saving the topology as a PDF. The actual scan output, MAC addresses, subnet and exported topology file were not supplied.

For the final submission, replace the illustrative host list with the exact addresses returned by the authorised local scan. Include MAC addresses only where actually observed and appropriate to disclose.

5. RISK ANALYSIS / IMPACT

#	Observation / Finding	Evidence supplied	Potential security significance	Indicative level
1	Web technology information exposed	WhatWeb identified WordPress 7.0.4 and	May help identify components for patch	Low

		WP Download Manager 3.3.58.	and configuration review.	
2	Domain resolves to an IP address	Nslookup returned 192.232.216.135.	Describes public-facing infrastructure; resolution alone is not a vulnerability.	Informational
3	HTTP response information available	cURL -I returned headers; /wp-json/ was noted.	May provide context for further authorised review.	Informational
4	WAF technology signature detected	WAFW00F identified ModSecurity (SpiderLabs).	Identifies a defensive technology; configuration and effectiveness were not tested.	Informational
5	DNS infrastructure information available	DNSRecon reviewed NS, MX, SPF/TXT and service-related records.	DNS records can reveal service relationships and should be reviewed for accuracy.	Low
6	Live local hosts listed in practical notes	Four example addresses supplied, not confirmed as actual LAN results.	An accurate inventory helps identify unknown or unmanaged devices; actual exposure remains unverified.	Pending

6. RECOMMENDATIONS

1. Review public technology disclosures. Maintain an accurate inventory of the CMS, plugins and web components, and assess whether version details are unnecessarily exposed.
2. Maintain supported software versions. Apply vendor security updates to WordPress, plugins, themes and server components through a controlled change process.
3. Review HTTP response headers and API exposure. Confirm that headers disclose no unnecessary information and that public API routes expose only intended data and functionality.
4. Maintain DNS hygiene. Periodically review authoritative name servers, mail records, TXT records and service records. Remove obsolete records and ensure email-authentication policies reflect the organisation's intended configuration.
5. Review WAF operation. Keep the identified WAF technology maintained and monitor its events. Validate its configuration through approved testing rather than assuming that detection of a WAF proves effective protection.
6. Maintain an authorised network inventory. Compare host-discovery results with approved asset records and investigate unknown devices through the organisation's normal IT process.
7. Document scope and evidence. Record the authorised target range, date and time, scan profile, relevant outputs and screenshots. Store reports and exported topology files securely.
8. Continue testing only within written scope. Obtain approval before moving from reconnaissance and host discovery into port or service testing, vulnerability validation or exploitation.

7. LIMITATIONS

This report covers reconnaissance and host discovery only. No exploitation, authentication testing, vulnerability scanning, denial-of-service testing or intrusive validation is reported.

The exact WHOIS output, complete DNSRecon records, HTTP header output, WAFW00F transcript, Zenmap scan file, actual local subnet, observed MAC addresses and topology PDF were not included in the material supplied for drafting. Accordingly, this report summarises the observations provided and does not reproduce unprovided raw evidence.

The software versions, IP address and WAF identification are time- and observation-dependent. Risk levels are indicative prioritisation labels, not results of a formal vulnerability assessment.

8. CONCLUSION

During Week 2 of the NetworkWalks Cybersecurity Internship, I completed practical exercises covering footprinting, reconnaissance and network discovery. The Kali Linux exercises demonstrated how WHOIS, WhatWeb, Nslookup, cURL, WAFW00F and DNSRecon can be used to gather and interpret publicly observable information about a domain and its web services.

The Zenmap exercise introduced local host discovery and the review of network topology. It reinforced the importance of identifying the correct local subnet, selecting an appropriate discovery profile and interpreting results in the context of an authorised asset inventory.

The exercises highlighted the difference between an observation and a confirmed vulnerability. A technology fingerprint, IP address, HTTP endpoint, DNS record or WAF signature may provide useful assessment context, but further authorised validation is needed before assigning a vulnerability finding.

Overall, the practical work strengthened my understanding of information gathering, network visibility, evidence documentation and responsible testing. Phases 3–5 remain in progress.

9. EVIDENCE

Supporting evidence: https://drive.google.com/file/d/1rO-8k5FS9b1LzM_D9rbr7fGXIsLmPPBL/view?usp=sharing